Department of Biostatistics,

Bioinformatics &

Biomathematics

Georgetown University Medical Center

Ph.D. Handbook

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Disclaimer: This information book is meant to be helpful. Although every attempt was made to ensure accuracy, some material may be out of date or potentially wrong. Thus, this document is not meant to represent a binding contract between faculty and students. In addition, students and faculty are recommended to review the Graduate School Bulletin and Biomedical Graduate Education (BGE) for information concerning all students.

PhD Program in Biostatistics

PART I: ADMISSIONS

Eligibility:

1) Bachelor's degree from an accredited college or university with a minimum GPA of 3.0 (out of 4.0), that includes:
   - Strong coursework in Multivariable Calculus and Linear Algebra (with grades of B or better)
   - Introductory course in Statistics or Biostatistics (with grades of B or better)

2) Master's degree from an accredited college or university, with a minimum GPA of 3.0 (out of 4.0) in a Quantitative Area:
   - Biostatistics, Statistics or Mathematics
   - **OR** Other Quantitative Areas (e.g., Economics, Computer Science, Biomedical Engineering, etc.), that includes strong coursework (with grades of B or better) in:
     - Statistical Inference (with grades of B or better)
     - Statistical Modeling (with grades of B or better)

3) GRE (optional). Minimum Quantitative GRE score of 155 (new score) or 700 (old score)

Admission Process:

The PhD Admission Committee consists of faculty assisted by a program coordinator. The Admissions Committee will review the applications, select candidates for interview, and make admission decisions and recommendations for financial support.

Financial Support:

Incoming Ph.D. students in Biostatistics may receive financial support. The award covers tuition, mandatory fees, student health insurance, and a stipend. The Department Education Committee will review the candidates recommended for financial support by the PhD Admission Committee and make the final decision on award of financial support.

Note:

- The application cycle for the Biostatistics PhD program is available in the Fall only.
- Admission of part-time students will be handled on a case-by-case basis.

PART II: CURRICULUM

Coursework is completed in the first 2 years of the program. Students complete 9 credits of required (core) courses, which provide a solid grounding in advanced statistical theory and methodology. Currently, all the elective courses below are required courses. An internship in the field of biostatistics is highly recommended. Introduction to Epidemiology MS course is also highly recommended to be taken in the first semester by all incoming PhD students who have not taken an epidemiology course before. Students must complete 21 credits of electives.

Students can select their electives (currently mostly in cognate courses) to tailor their training toward research interests. Classes are held following the Georgetown University operating status. Thesis Research (BIST 999) typically begins in the 3rd year. Research should constitute a significant original contribution to the field of Biostatistics.

Below is a typical course schedule:

Year 1 **Fall Semester**		**Spring Semester**	
Course Name	**Credit(s)**	**Course Name**	**Credit(s)**
BIST 6500: Department Seminar	0	BIST 6500: Department Seminar	0
BIST 7610: Probability and Large Sample Theory	3	BIST 7615: Advanced Statistical Inference	3
BIST 7625: Statistical Computing*	3	BIST 7630: Bayesian Inference*	3
BIST 7620: Generalized Linear Models	3	BIST 7635: Longitudinal Data Analysis*	3
Total 9 Credits		**Total 9 Credits**	

Year 2 **Fall Semester**		**Spring Semester**	
Course Name	**Credit(s)**	**Course Name**	**Credit(s)**
BIST 6500: Department Seminar	0	BIST 6500: Department Seminar	0
BIST 7640: Causal Inference*	3	BIST 7655: Statistical Genetics*	3
BIST 7645: Advanced Survival Analysis*	3	BIST 7660: Deep Learning & Artificial Intelligence*	3
BIST 7650: Semiparametric Inference*	3	BIST 7665: Ethics of Biostatistics and Collaboration*	1
Cognate Requirement[c]	3	Cognate Requirement[c]	3
Total 12 Credits		**Total 10 Credits**	

Year 3 + **Fall or Spring Semesters**	
Course Name	**Credit(s)**
BIST 6500: Department Seminar	0
BIST 999: Thesis Research	12
Total 12 Credits	

**** Biostatistics Electives (Required Courses)***
***** Bioinformatics Requirement*** [c] ***Cognate Requirements*** *(may be taken anytime from Year 2+)*

Cognate requirement. Please discuss with your academic advisors and Dr. Kepher Makambi the selection of courses for satisfying this requirement.

Other requirements (0 credit)

Research ethics requirement can be fulfilled by completing the CITI Program online training course in Human Research, Group 1 Biomedical research investigators and key personnel at Stage 1 - Basic Course and Conflict of Interest.

PART III: COMPREHENSIVE EXAMINATION

Students who wish to pursue the PhD degree must pass the comprehensive written exam. The exam covers material in probability, inference and other core courses. The exam consists of 2 parts: (1) Theory , and (2) Methods and Application. Each part of the exam will take 4 hours. Comprehensive examinations are administered in accordance with Georgetown University's operating status.

The Theory part will be at the level of the Casella and Berger textbook "Statistical Inference." The Methods- Application part will be based on the key course content of BIST 7620 - Generalized Linear Models.

Students in the PhD program must attempt the comprehensive exam in the summer after completing the core courses. For example, students can make up for any deficiencies before taking the examination by enrolling in appropriate master's-level courses as needed. Every PhD student is allowed to attempt the exam twice. If a student does not pass on the first attempt, namely, if the student does not pass any one of the two parts, the student is considered not to pass the comprehensive exam at this first attempt. A retake of the part(s) of the exam (Theory and/or Methods/Application) that the student does not pass on the first attempt is permitted and must take place the next time the exam is offered. If the student passes the relevant part(s) on the second attempt, the student passes the comprehensive exam. Failure in the second attempt will lead to discontinuation from the PhD program.

The duration of each part will be determined by the comprehensive examination committee (CEC). The comprehensive examination committee, formed in 2018, currently consists of a chair and three members who have instructed the three core courses. The CEC will be responsible for preparing, administering, and grading the examination. Before the exam is administered, the committee chair will call and hold a committee meeting to create the exam structure according to the questions provided by committee members. The committee chair will review the exam questions and provide input to members. After the exam is completed, the Core Faculty who taught the courses will score the exams, and the scores of each part recorded separately in the faculty member section of the graduate's Exam Report Form. The committee will discuss the overall pass/fail score, and then the CEC chair will fill out the Final Results portion of the Exam Report Form and forward it to the PhD Program Director for signature, and send it to Graduate School.

PART IV: THESIS RESEARCH

During the third and fourth years of the program, students devote most of their time to thesis research in preparation of the doctoral thesis. During these years, students are expected to attend and participate in seminars/colloquia sponsored by DBBB. . While carrying out thesis research, the student may, with the instructor's and mentor's permission, also sit in on advanced courses either in statistics or any subject of particular relevance to their thesis research or long-term career interests.

Typical Timetable for Thesis Research	Time after passing the Comprehensive Exam (typical semester from the start of the program)
Mentor and/co-mentor are selected, thesis research can start	By the following year (3rd or 4th semester). Encouraged to start as early as possible
Grad School Required Thesis Proposal Form and Thesis Committee Selection	By the 5th Semester. This can be combined with the first thesis committee progress report
Thesis Committee review meetings and Progress Report	At least annually by May 31. The committee is encouraged to meet twice annually, once each semester. See the sample schedule under the Thesis Committee
Oral Qualifying Exam	By 6th or 7th Semester. This exam should be aimed midway between the start and the defense of the thesis, or 6 months before the final defense, or in approved special cases, at least 6 weeks before the final defense, to meet graduate school deadlines.
Thesis Defense	2 years. March-August (8th Semester)

Thesis Advisor/ Mentor Selection

When a student has passed Biostatistics PhD Comprehensive Written Examination, the next step is to select a faculty mentor(s) and thesis research project moving forward.
Criteria for a good match include (1) mutual agreement between the student and advisor on a research project, and (2) an excellent research environment in the mentor's research team. It would be ideal if funding in the thesis research area is available. Typically, a co-mentor(s) in an area of expertise complementary to the primary biostatistics mentor may also be chosen. For example, a thesis on innovative methods for analyzing next-generation sequences and/or modeling tumor heterogeneity may need a primary biostatistics mentor and biological and mathematical co-mentors. It is also possible for a candidate who is especially strong in mathematics and has a strong interest in mathematical approaches to biological problems to work with a mentor from the Department of Mathematics and Statistics and a co-mentor(s) from DBBB. Mentor assignments will be monitored to ensure that each advisor has adequate time for mentoring the PhD student. The Education Committee will assist any student with necessary changes in mentorship. If a chosen mentor is not a member of the Georgetown faculty, a member of the Georgetown faculty must serve as co-mentor to regularly monitor the candidate's research and progress toward the degree and report to the oversight committee. **So, mentors in biostatistics or quantitative research subspecialty areas closely related to biostatistics, e.g., differential equations, artificial intelligence, network and –omics analysis, and classification algorithms, are designated as primary mentors, those in the complementary field of tumor biology and oncology as co-mentors**. Mentors are expected to be chosen by May 1 of the following year after a student passes the Comprehensive Written Exam.

Once the mentor and student reach an agreement on the selection, they will email the DGS and the Program Coordinator to confirm the selection. For first-time thesis mentors (i.e., someone who has not yet graduated as a Ph.D. student), a co-mentor who has is encouraged.

GRA and Responsibilities

All student assistantships are governed by GSAS and BGE policies. Unless a student is supported on a university fellowship, e.g., the first two years of the BGE scholarship, the stipend will typically have service obligations. In the Biostatistics Program, the service component in a BGE-awarded stipend package takes multiple forms: (1) working as a Master level statistician on collaborative research contracts/grants; (2) graduate research assistant on a specific grant which ties with the thesis research topic; (3) NIH partnership jointly with GU and NIH mentors; and (4) NIH awarded predoctoral dissertation fellowship grants (e.g., F31) that supports the PI (the student here) with a stipend to work on her or his dissertation.

Service stipends with collaborative research service components are generally determined by the thesis mentor and/or project faculty supervisor(s) in conjunction with the PhD Program Director. The thesis mentor should discuss the responsibilities of the GRA and its relation to academic work (dissertation research) with the mentee. Collaborative research is integral to the training of a Ph.D. in biostatistics, but the service component can be, and many times is, independent of the thesis research. So, the thesis mentor, project supervisor, and the student need to recognize that the students may have dual responsibilities that need to be managed effectively. The thesis mentor and project faculty supervisor(s) are expected to facilitate the accomplishment of both thesis research and the collaborative research service component. It is important to recognize the service component in our case is collaborative and consulting research project work, which provides the student opportunities to learn how biostatistics is practiced in real life and is of obvious importance to the success of students, e.g., in placement. Therefore, a student is encouraged to take a biostatistics dry-lab rotation if she or he feels some additional training is needed to take on data analysis projects. In addition, biostatistics method and collaborative research often are interrelated, e.g., the collaborative project motivates novel statistical method developments which can then be applied to help the collaborative research projects. It would be ideal and the goal is to strive to make the stipend work overlaps with thesis research as much as possible, and ultimately thesis research funded on a grant such as an NIH T32, and/or NIH F31 grant that the student will apply with help from mentors, and a research grant.

Within the biostatistics shared resource (BSR) budget approved by GUMC, which may have 1-2 stipend positions, the education committee will recommend to the BSR director students who will be the best fit based on experience and academic performance e.g., GPA as a guide or dry-lab rotation experience. The BSR director in conjunction education committee will select the candidate based on academic performance, e.g., GPA as a guide, and the needs of the BSR. The BSR is expected to recover the funding through charge-back, namely, invoicing the hours the student has worked. All faculty can and are highly encouraged to bring in collaborative or consulting projects.

Thesis Committee

After successfully passing the written comprehensive exam and identifying a thesis mentor, a student will form a thesis committee in consultation with the mentor(s). This committee will guide the student's subsequent progress toward the degree. It consists of at least four members, all of whom must hold the rank of assistant professor or above. One of the committee members must be a faculty member from another discipline other than those with a primary DBBB appointment. The committee is selected by mutual agreement between the student and the advisor and typically formed by the end of the first semester into the thesis stage. The mentor usually chairs the committee. The Thesis Committee will be confirmed by completing the graduate school Thesis Proposal Form.

The thesis committee meets at least twice per year by May 31 to review student progress and to provide a report (see Thesis Committee Meeting/Progress Report form) to the program oversight committee body, namely, the Education Committee, and acknowledged by the student after every thesis committee meeting. All thesis committee members are expected to attend in person (encouraged) or via Zoom. In any case, a quorum of three or more thesis committee members must be maintained. For oral qualifying and final oral (thesis defense) examinations, a quorum of at least three members in person has to be maintained. Students will present their thesis progress using appropriate written or visual aids and obtain feedback and suggestions to enhance their research productivity. Based on data from 6 students who have completed our program, a sample meeting schedule during the thesis research period is below, assuming the thesis research period is two years.

Thesis Research Semester1/Mtg 1	Thesis Research Semester 2/Mtg 2	Thesis Research Semester3/Mtg 3/Oral Qualifying Exam	Thesis Research Semester 4/Mtg 4	Final Thesis Defense
Kickoff meeting: Thesis committee is formed by completing GSAS thesis proposal form. Comments to Committee Meeting /Progress Report made	Progress Report/ Committee Comments	The mentor and thesis committee decides if the additional meeting is needed or hold the oral qualifying exam (see Part V description). This exam is expected at the midpoint between the start and finish of the thesis, or 6 months before the final defense, or in approved special cases, at least 6 weeks before the final defense, to meet graduate	Progress Report/ Committee Comments. A draft of the thesis may be sent to committee for input. Progress Report provides comment on readiness on thesis defense	Pay attention to timeline and forms. Work with Program Coordinator for signatures.

		school deadlines.		

The thesis is based on independent research by the candidate and should constitute a significant original contribution to the field of biostatistics, involving the development and evaluation of biostatistical methodology although the methods are motivated by biomedical research problems. Software development and computational techniques for novel statistical methods is also important aspect of scholarly work. A thesis may consist of three separate (maybe related) papers that are judged to be of publishable quality or an in-depth treatise of a topic that may look at various facets of a problem and may not easily be subdivided into a given number of separate publishable papers. So, a non-straightforward statistical analysis of a biomedical research data set alone does not qualify as a biostatistics PhD thesis. The award of Ph.D. does not require the publication of the dissertation work at the time of thesis defense, but it is highly encouraged that students to couple the two, e.g., to be more competitive in placement.

We encourage all of our trainees to complete an individual development plan (IDP) and review it semi-annually with their thesis mentor. If progress is deemed unsatisfactory by the mentor and thesis committee, the mentor must first issue a warning, usually in writing, stating the problems with the student's progress. Next, another Thesis Committee meeting is called to discuss short term goals for the student to accomplish in the next three to four months. The student is given the opportunity to reach these goals in the allotted time, and then a follow-up meeting of the Thesis Committee is called. At this time, if the student's progress is still deemed unsatisfactory, the student may be terminated from the project and the relationship. In the unlikely event that the student cannot find another mentor willing to sponsor their thesis research, the student may be asked to leave the program with an ABD acknowledgment. Students who are dismissed from the Ph.D. program will not be considered for re- admittance

Likewise, a student may feel that their thesis research is not progressing due to underlying issues with the thesis mentor and their research. The student should openly communicate this to the mentor (in the presence of the Director of Graduate Studies if needed) in an effort to reach a mutual agreement. If the student and mentor cannot come to an understanding regarding the research project, then the student is responsible for finding another mentor to fund their research before switching mentors. Again, if the student cannot find another laboratory to sponsor their thesis research, the student may be asked to leave the program.

Thesis Proposal

The thesis proposal document (form) is required by the Graduate School and is approximately two to eight pages, not including figures. The purpose of the document is to serve as a focus for discussion at thesis committee meetings and to allow the student and mentor to formulate a viable thesis research plan. This should be completed by the first thesis committee meeting.

Thesis Committee Meetings

Every student's first Thesis Committee meeting must occur shortly after the two full semesters in thesis, typically by the end of the second semester. After that, they must meet a minimum of twice per year, or more frequently as determined by the committee. Meetings should typically follow the student's annual seminar presentation. The presentation to the committee should be relatively informal and does not constitute an examination. It is highly recommended that the student circulates the Thesis Proposal, along with other supporting materials (scientific

abstracts), to all members of the committee prior to the first meeting.

The thesis proposal format should approximate the following:

I. Introduction (one to two pages): review of literature supporting proposed project
II. Objectives and specific aims (up to one page)
III. Methods and experimental design (one to four pages): prose or outline as the student prefers
IV. Preliminary results: tables and figures with legends
V. Proposed experiments for the next year: outline

Following acceptance by the thesis committee the Thesis Proposal should be submitted to the Graduate School using their designated form.

As this is one of the thesis committee meetings, the committee may fill out the progress report based on the same meeting especially when there are suggestions that need to be conveyed to the student. The mentor and student may choose to combine the open session with a colloquium as PHD could report their work in colloquiums anyway.

At each committee meeting, students are urged to prepare a brief outline with data, figures and proposed future experiments and/or studies. Any manuscripts, either written or in preparation, should be given to the committee in advance. The student will be evaluated at each Thesis Committee meeting using the form designated for this purpose. The reports will be placed in the student's file, which is maintained by the department's Program Coordinator. Failure to progress during thesis research, as evaluated by the thesis committee may be grounds for dismissal from the program.

PART V: ORAL QUALIFYING EXAMINATION

The **Oral Qualifying Examination** will be taken before advancement to candidacy and after successfully passing the comprehensive examination, having identified a thesis topic, thesis mentor(s)/advisor(s), and a thesis Committee. It is held following the Georgetown University operating status. The exam proceeds with the student presenting the thesis proposal to the thesis committee, followed by a question and answer period covering the special research topic and coursework related to the thesis topic. The committee will determine the student's readiness to pursue and complete the proposed research and the appropriateness of specific problems for thesis research. Preparation for the exam will be done by working closely with the thesis mentor. Although progress in thesis research varies, this exam usually takes place one year into the thesis research or around the midpoint between the start and finish of the dissertation, and no later than 6 months, or at least 4-6 weeks, before the final defense, in order to meet the graduate school deadline.

A student who fails the Oral Qualifying Examination may be given one more chance to retake it. The program director and/or Thesis Committee will specify the time for the reexamination.

Advancement to candidacy will be granted when it is determined that the student has completed all the requirements for the doctoral degree except the thesis/thesis.

PART VI: FINAL ORAL EXAMINATION: Thesis Defense

To prepare for defense, a student will need to get to a defensible draft of their theses. If the mentor deems the thesis is ready, the student will send that draft thesis to the entire thesis committee to review and sign off the form on whether the thesis is ready for defense and let PhD Program Coordinator know in order to send the graduate school thesis ballot on thesis readiness to the mentor and the rest of the thesis committee members to sign through DocuSign. Thesis defense is held in accordance with the Georgetown University operating status. The student is expected to follow the graduate school CHECKLIST FOR DISSERTATION WRITERS. Upon completion of the thesis defense, the Ph.D. program coordinator will send the committee the thesis defense exam form (ballot) to assign. The PhD candidate will then produce a final copy of the thesis. The committee members will need to review if it is satisfactory and to endorse it by signing the Thesis Cover Sheet form and having it sent to the dean of the graduate school to review and sign.

Defense of the thesis will constitute the final examination for the PhD degree. The thesis examination committee is typically the thesis examination committee but is chaired by a committee member other than the mentor. The thesis committee will name the chair and send the name of the chair to the PhD program coordinator for record. The doctoral thesis defense will have an open and a closed session. The open session includes a public presentation by the candidate which is announced to our regular department seminar and colloquium community, in addition to the GSAS calendar announcement to all members of the Georgetown academic community. Therefore, a title and abstract shall be distributed and all those interested can attend. More specifically, the thesis will be made available to DBBB and attendees if requested in the Box when the announcement makes it to graduate school. The closed session usually starts with a period during which the committee alone may ask the candidate questions. Then the rest of the session will be a closed meeting of the committee during which it deliberates and decides whether or not the defense was successful.

The appointed exam committee chair will call the defense to order. The chair will ask the mentor to introduce the student, and call on the candidate for presentation of the thesis, typically a 30-45-minute presentation and then call on the general audience, followed by committee members, for questions (this may be varied). After the public presentation, the committee will have time for closed session deliberations. Pass or no pass is determined by a majority vote of the thesis committee.

If the student passes the defense, the student and/or mentor will work with the PhD Program Coordinator to have the required forms (exam ballot and PhD dissertation coversheet) signed. Given the GSAS deadlines for graduation paperwork, it is highly recommended that the defense be held no later than July.

It is a must that students review the graduate school website on thesis research, defense and submission process at https://grad.georgetown.edu/dissertation-doctoral-project-and-thesis-information/

The related forms can be found at
https://grad.georgetown.edu/academics/current-students/academic-forms/#theses-forms
Other websites of interest include
https://grad.georgetown.edu/policies/
and the Graduate Bulletin https://sites.google.com/a/georgetown.edu/gsas-graduate-bulletin